Genome-wide fusion-hotspot analysis of msk_impact_50k_2026

Abstract

This manuscript synthesizes a genome-wide fusion-hotspot cohort scan of msk_impact_50k_2026: 3919 genes carried at least one structural-variant record in the cohort, of which 544 passed the >= 5-distinct-patient recurrence gate. All 544 gated genes were attempted: 10 using hand-curated gene configs and 528 auto-configured, with 4 gated-in gene(s) left unresolvable; 538 of the 544 attempted genes were successfully analyzed with the full registered algorithm suite. 17 genes reached genome-wide Benjamini-Hochberg FDR significance (q < 0.05): BRAF (179 events, 92.6% in-frame, 91.6% domain-retained, Fisher p=0.0371318, q=0.000624279); CD74 (67 events, 68.4% in-frame, 89.4% domain-retained, Fisher p=0.270083, q=0.000624279); EGFR (55 events, 70.0% in-frame, 74.5% domain-retained, Fisher p=0.018443, q=0.000624279); EML4 (223 events, 97.5% in-frame, 53.4% domain-retained, Fisher p=0.999931, q=0.000624279); ETV6 (90 events, 81.0% in-frame, 77.3% domain-retained, Fisher p=0.000290182, q=0.000624279); FGFR2 (136 events, 93.1% in-frame, 85.8% domain-retained, Fisher p=0.00226743, q=0.000624279); FGFR3 (152 events, 97.4% in-frame, 94.0% domain-retained, Fisher p=0.0194679, q=0.000624279); KIAA1549 (44 events, 97.6% in-frame, 50.0% domain-retained, Fisher p=0.927235, q=0.000624279); KIF5B (96 events, 97.7% in-frame, 96.9% domain-retained, Fisher p=0.307781, q=0.000624279); PRKACA (35 events, 96.0% in-frame, 96.7% domain-retained, Fisher p=0.214286, q=0.000624279); RET (194 events, 85.4% in-frame, 92.3% domain-retained, Fisher p=0.00256613, q=0.000624279); TMPRSS2 (867 events, 33.3% in-frame, 6.2% domain-retained, Fisher p=0.0849513, q=0.000624279); ATF1 (30 events, 84.0% in-frame, 90.0% domain-retained, Fisher p=0.704212, q=0.00109249); TPM3 (14 events, 100.0% in-frame, 64.3% domain-retained, Fisher p=1, q=0.00109249); ROS1 (122 events, 72.7% in-frame, 90.7% domain-retained, Fisher p=0.186019, q=0.0019422); NAB2 (72 events, 71.4% in-frame, 69.0% domain-retained, Fisher p=0.179862, q=0.00919908); ALK (272 events, 97.4% in-frame, 96.3% domain-retained, Fisher p=0.00057792, q=0.016646). 15 additional genes form a highly ranked non-FDR-significant tier flagged for targeted follow-up (see Honorable mentions, below).

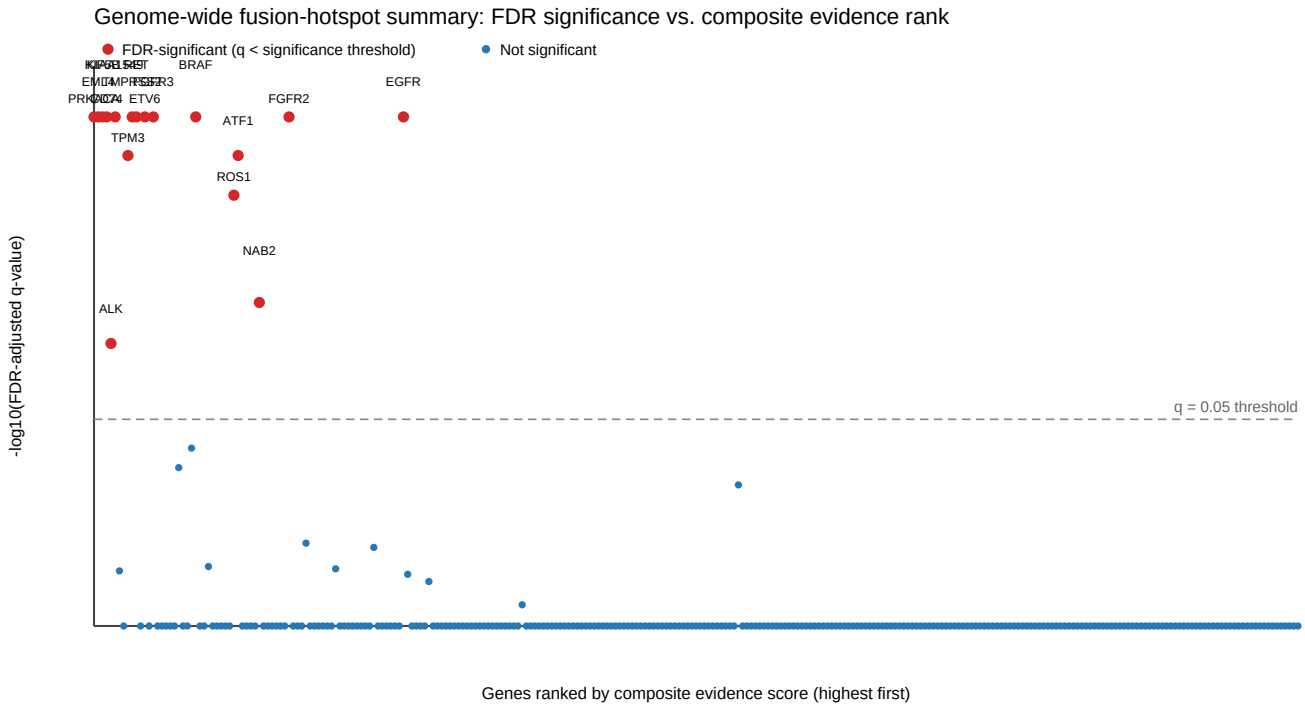
Methods

Structural-variant records were retrieved from the msk_impact_50k_2026 cBioPortal study and gated to genes with at least 5 distinct patient(s) carrying a structural-variant record (544 of 3919 genes passed the gate). All 544 gated genes were attempted: 10 using hand-curated gene configs and 528 auto-configured from Genome Nexus canonical-transcript/Pfam-domain data, with 4 gated-in gene(s) left unresolvable; 538 of the 544 attempted genes were successfully analyzed. Each successfully analyzed gene was run through the algorithm suite recorded for this scan (composite_score, confidence_stats, cutpoint_detection, domain_disruption, domain_retention, exon_retention, expression_association, frequency, genomic_position_recurrence, joint_partner, junction_concentration, mechanistic_interpretation, mutation_cooccurrence, window_detection). Domain-retention and domain-disruption significance were assessed per gene with Fisher's exact test and a breakpoint-position permutation test; the resulting p-values across the 285 genes that produced at least one computable p-value were jointly corrected with Benjamini-Hochberg false-discovery-rate correction at q < 0.05.

Results

Genome-wide summary

Genome-wide summary of 285 scanned genes with an FDR-adjusted q-value, ranked left-to-right by composite evidence score; the dashed line marks the q=0.05 significance threshold, with 17 genes above it.


Genome-wide fusion-hotspot summary: FDR significance vs. composite evidence rank
FDR-significant (q < significance threshold)
Not significant
BRAF
EGFR
FGFR2
ETV6
ATF1
TPM3
ROS1
NAB2
ALK
q = 0.05 threshold
-log10(FDR-adjusted q-value)
Genes ranked by composite evidence score (highest first)

FDR-significant and honorable-mention genes

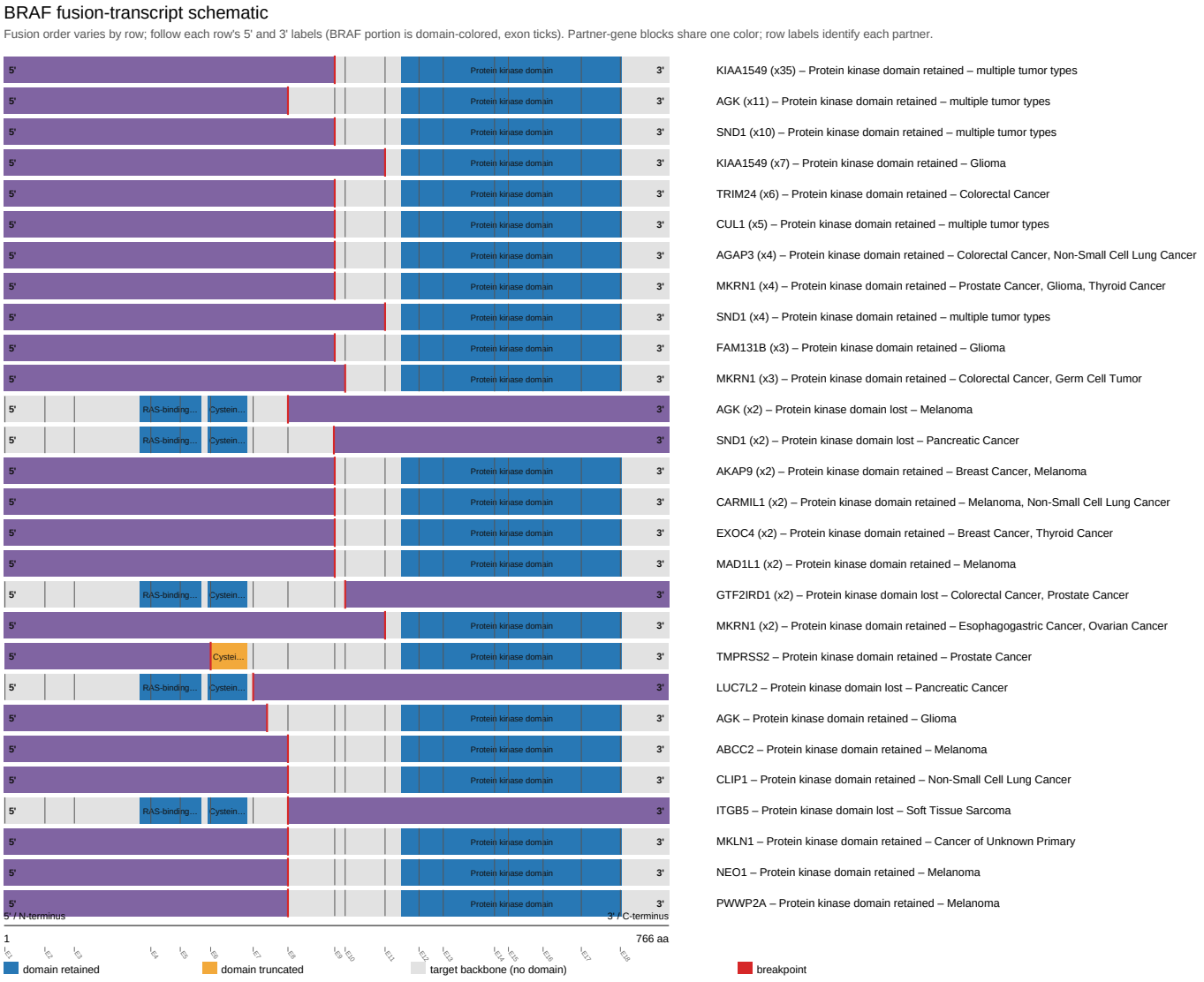
Gene	Tier	N events	In-frame%	Domain-ret.%	Fisher p	Best q
BRAF	FDR-significant	179	92.64	91.57	0.03713	0.0006243
CD74	FDR-significant	67	68.42	89.39	0.2701	0.0006243
EGFR	FDR-significant	55	70	74.55	0.01844	0.0006243
EML4	FDR-significant	223	97.51	53.36	0.9999	0.0006243
ETV6	FDR-significant	90	81.01	77.27	0.0002902	0.0006243
FGFR2	FDR-significant	136	93.1	85.82	0.002267	0.0006243
FGFR3	FDR-significant	152	97.37	94.04	0.01947	0.0006243
KIAA1549	FDR-significant	44	97.56	50	0.9272	0.0006243
KIF5B	FDR-significant	96	97.7	96.88	0.3078	0.0006243
PRKACA	FDR-significant	35	96	96.67	0.2143	0.0006243
RET	FDR-significant	194	85.38	92.27	0.002566	0.0006243
TMPRSS2	FDR-significant	867	33.33	6.243	0.08495	0.0006243
ATF1	FDR-significant	30	84	90	0.7042	0.001092
TPM3	FDR-significant	14	100	64.29	1	0.001092
ROS1	FDR-significant	122	72.73	90.68	0.186	0.001942
NAB2	FDR-significant	72	71.43	69.01	0.1799	0.009199
ALK	FDR-significant	272	97.41	96.31	0.0005779	0.01665
CDKN2B	Honorable mention	12	66.67	20	0.02222	0.5249
NTRK1	Honorable mention	78	72.73	84.21	0.05145	0.3201
NTRK3	Honorable mention	58	98.21	96.49	0.05556	0.07592
IKBKE	Honorable mention	12	50	36.36	0.06667	1
FLI1	Honorable mention	118	68.22	94.02	0.07688	1
RTEL1	Honorable mention	24	50	37.5	0.1167	1
KDM5A	Honorable mention	10	50	55.56	0.119	1
FH	Honorable mention	16	16.67	13.33	0.1429	1
CDK12	Honorable mention	43	50	39.02	0.1734	1
POLD1	Honorable mention	16	33.33	46.15	0.1923	0.7353
AGO2	Honorable mention	23	46.15	31.82	0.2252	1
NRG1	Honorable mention	15	100	86.67	0.2273	1
MTAP	Honorable mention	4	33.33	25	0.25	1
RPS6KA4	Honorable mention	13	62.5	58.33	0.2619	1
PLK2	Honorable mention	7	50	57.14	0.2857	1

Gene highlights

BRAF (FDR-significant, Curated gene config)

BRAF was analyzed across 179 fusion events, 92.6% in-frame and 91.6% domain-retained. Domain-retention Fisher's exact test on BRAF's own fusion events alone gives p=0.0371318 (statistically significant at alpha=0.05). The Protein kinase domain appears to be required for retention. Among 285 genes scanned genome-wide in this cohort run, BRAF's Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance).

Reused from the BRAF individual gene report (fusion_schematic.svg).



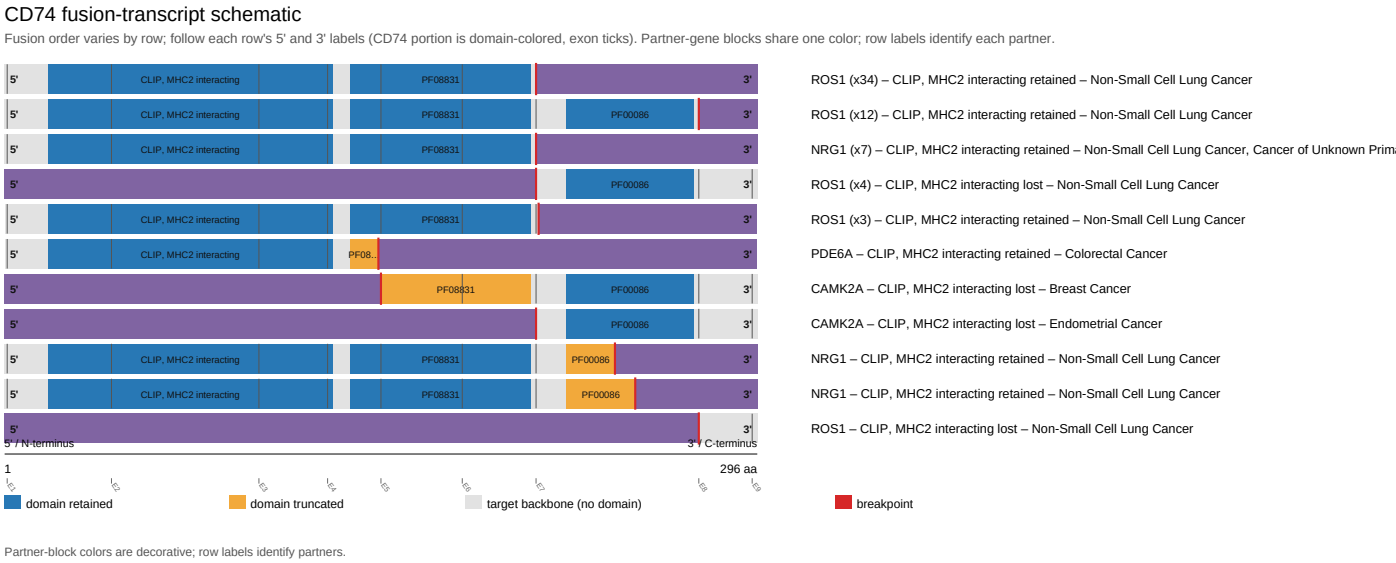
Full per-gene detail: gene_reports/braf/report.md

CD74 (FDR-significant)

CD74 was analyzed across 67 fusion events, 68.4% in-frame and 89.4% domain-retained. Domain-retention Fisher's exact test on CD74's own fusion events alone gives p=0.270083 (not statistically significant at

alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, CD74's Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance).

Reused from the CD74 individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/cd74/report.md

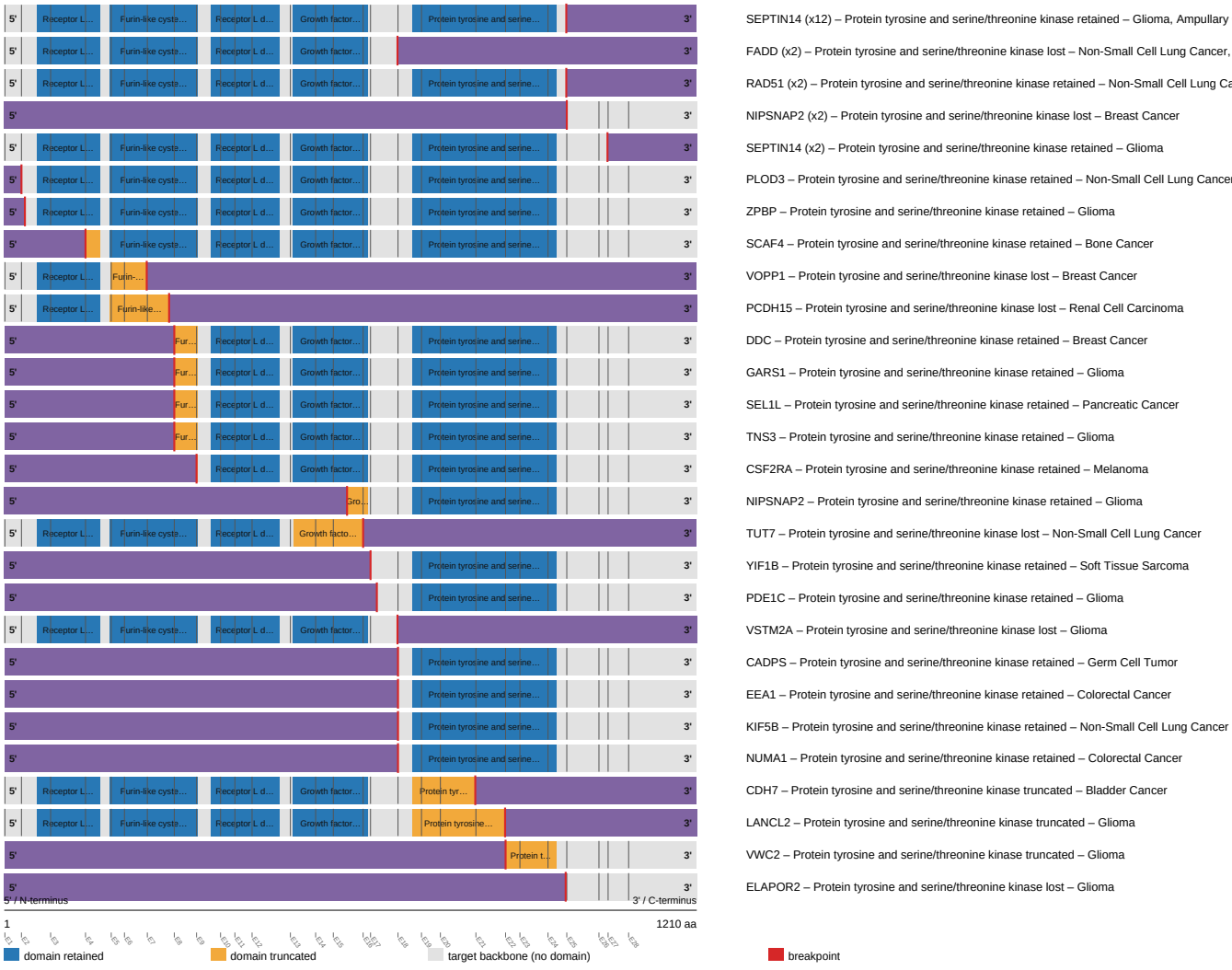
EGFR (FDR-significant)

EGFR was analyzed across 55 fusion events, 70.0% in-frame and 74.5% domain-retained. Domain-retention Fisher's exact test on EGFR's own fusion events alone gives p=0.018443 (statistically significant at alpha=0.05). The Protein tyrosine and serine/threonine kinase appears to be required for retention. Among 285 genes scanned genome-wide in this cohort run, EGFR's Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance).

Reused from the EGFR individual gene report (fusion_schematic.svg).

EGFR fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (EGFR portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

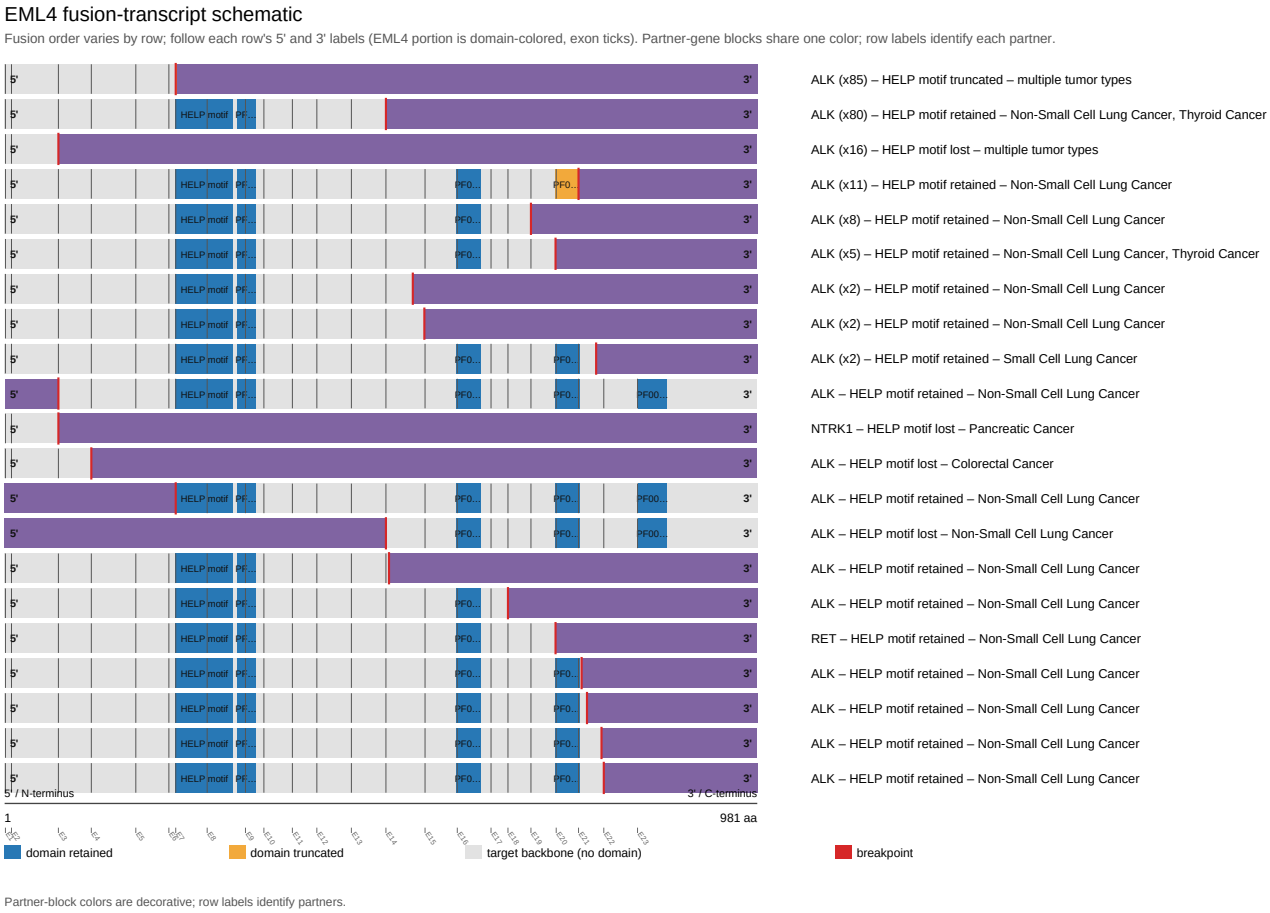
Showing the top 28 of 40 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/egfr/report.md

EML4 (FDR-significant)

EML4 was analyzed across 223 fusion events, 97.5% in-frame and 53.4% domain-retained. Domain-retention Fisher's exact test on EML4's own fusion events alone gives p=0.999931 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, EML4's Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance).

Reused from the EML4 individual gene report (fusion_schematic.svg).

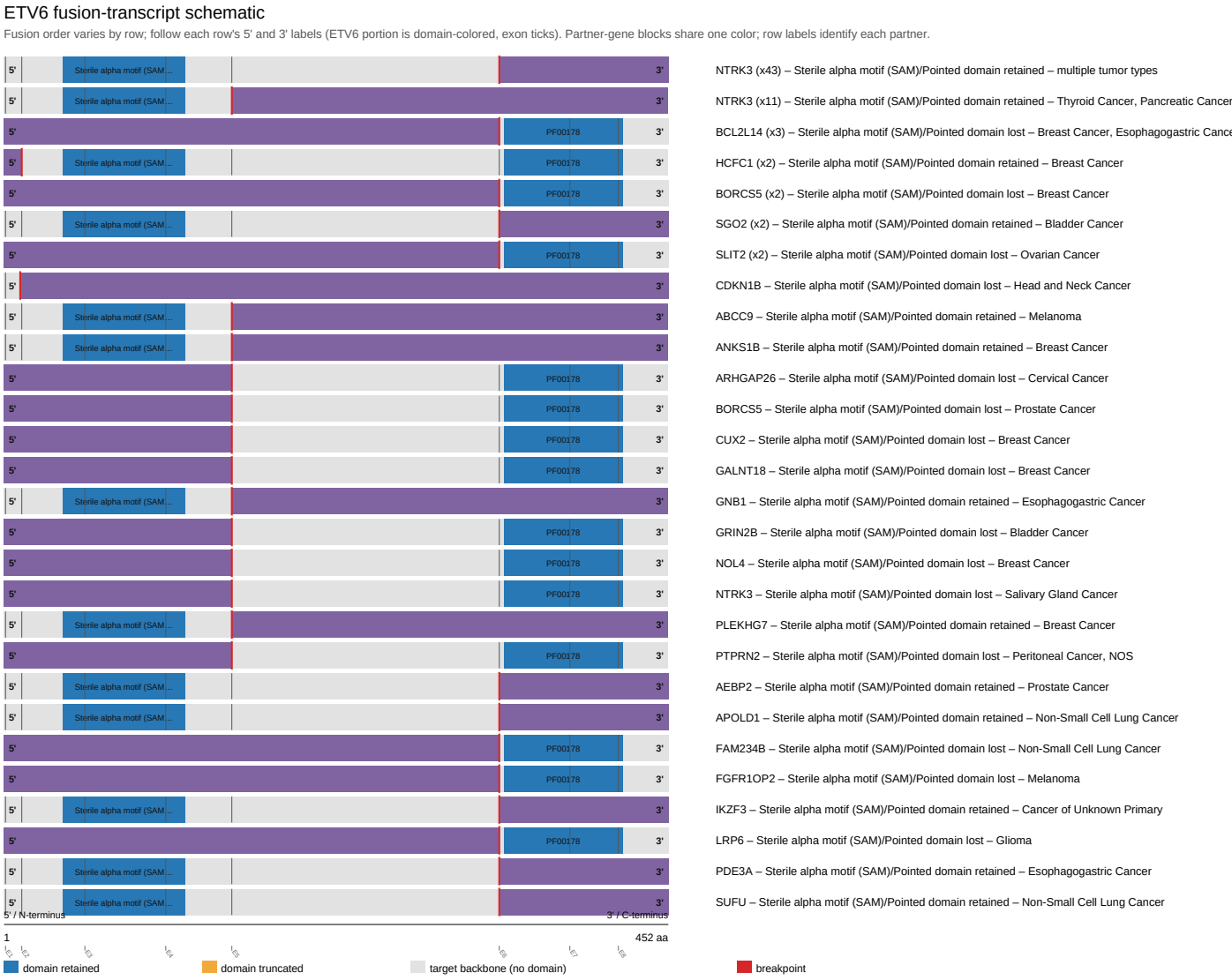


Full per-gene detail: gene_reports/eml4/report.md

ETV6 (FDR-significant, Curated gene config)

ETV6 was analyzed across 90 fusion events, 81.0% in-frame and 77.3% domain-retained. Domain-retention Fisher's exact test on ETV6's own fusion events alone gives p=0.000290182 (statistically significant at alpha=0.05). The Sterile alpha motif (SAM)/Pointed domain appears to be required for retention. Among 285 genes scanned genome-wide in this cohort run, ETV6's Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance).

Reused from the ETV6 individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/etv6/report.md

FGFR2 (FDR-significant, Curated gene config)

FGFR2 was analyzed across 136 fusion events, 93.1% in-frame and 85.8% domain-retained. Domain-retention Fisher's exact test on FGFR2's own fusion events alone gives p=0.00226743 (statistically significant at alpha=0.05). The Protein kinase domain appears to be required for retention. Among 285 genes scanned genome-wide in this cohort run, FGFR2's Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance).

Reused from the FGFR2 individual gene report (fusion_schematic.svg).

FGFR2 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (FGFR2 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 84 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/fgfr2/report.md

FGFR3 (FDR-significant)

FGFR3 was analyzed across 152 fusion events, 97.4% in-frame and 94.0% domain-retained. Domain-retention Fisher's exact test on FGFR3's own fusion events alone gives p=0.0194679 (statistically significant at alpha=0.05). The Protein tyrosine and serine/threonine kinase appears to be required for retention. Among 285 genes scanned genome-wide in this cohort run, FGFR3's Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance).

Reused from the FGFR3 individual gene report (fusion_schematic.svg).

FGFR3 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (FGFR3 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

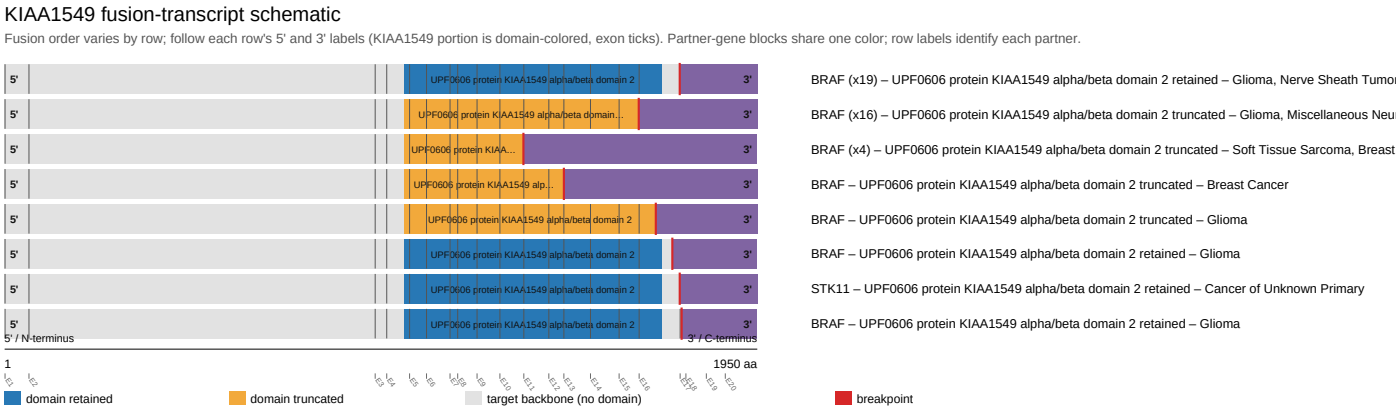
Showing the top 28 of 51 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/fgfr3/report.md

KIAA1549 (FDR-significant)

KIAA1549 was analyzed across 44 fusion events, 97.6% in-frame and 50.0% domain-retained. Domain-retention Fisher's exact test on KIAA1549's own fusion events alone gives p=0.927235 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, KIAA1549's Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance).

Reused from the KIAA1549 individual gene report (fusion_schematic.svg).

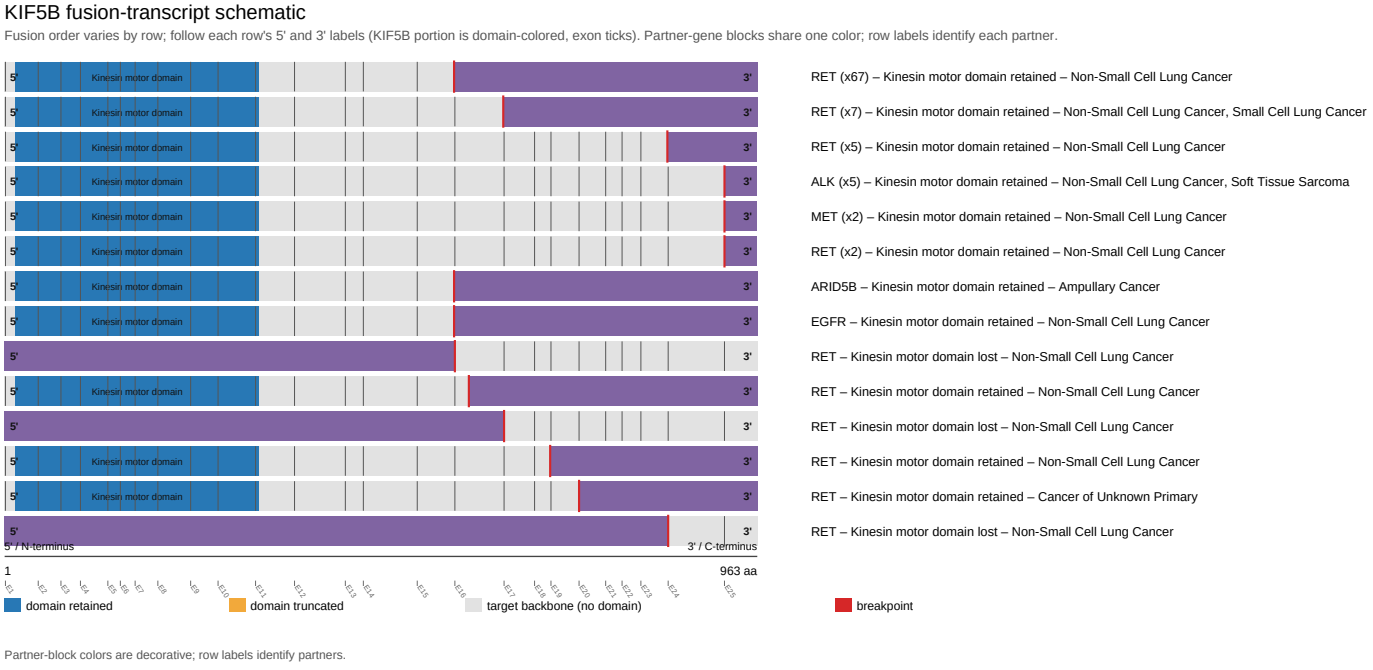


Full per-gene detail: gene_reports/kiaa1549/report.md

KIF5B (FDR-significant)

KIF5B was analyzed across 96 fusion events, 97.7% in-frame and 96.9% domain-retained. Domain-retention Fisher's exact test on KIF5B's own fusion events alone gives p=0.307781 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, KIF5B's Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance).

Reused from the KIF5B individual gene report (fusion_schematic.svg).



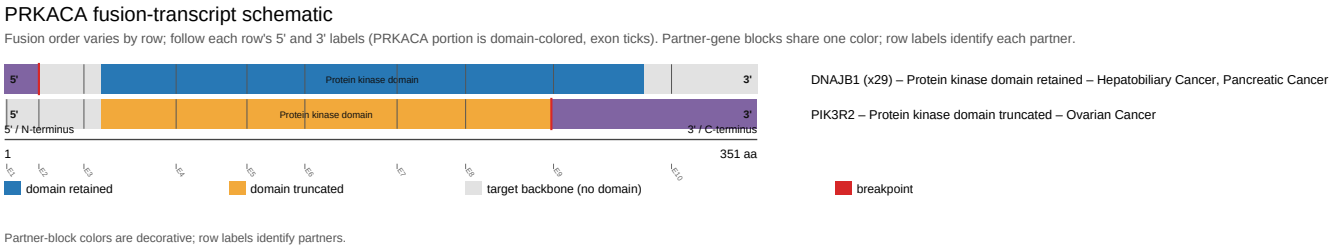
Full per-gene detail: gene_reports/kif5b/report.md

PRKACA (FDR-significant)

PRKACA was analyzed across 35 fusion events, 96.0% in-frame and 96.7% domain-retained. Domain-retention Fisher's exact test on PRKACA's own fusion events alone gives p=0.214286 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, PRKACA's

Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance).

Reused from the PRKACA individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/prkaca/report.md

RET (FDR-significant, Curated gene config)

RET was analyzed across 194 fusion events, 85.4% in-frame and 92.3% domain-retained. Domain-retention Fisher's exact test on RET's own fusion events alone gives p=0.00256613 (statistically significant at alpha=0.05). The Protein kinase domain appears to be required for retention. Among 285 genes scanned genome-wide in this cohort run, RET's Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance). The Cadherin domain appears to require loss or disruption rather than retention.

Reused from the RET individual gene report (fusion_schematic.svg).

RET fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (RET portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



KIF5B (x78) – Protein kinase domain retained – Non-Small Cell Lung Cancer, Cancer of Unknown Prima
CCDC6 (x45) – Protein kinase domain retained – multiple tumor types
NCOA4 (x16) – Protein kinase domain retained – multiple tumor types
SPECC1L (x4) – Protein kinase domain retained – Soft Tissue Sarcoma, Breast Cancer, Renal Cell Car
ERCC6 (x3) – Protein kinase domain lost – Non-Small Cell Lung Cancer
TIMM23B (x3) – Protein kinase domain retained – Non-Small Cell Lung Cancer, Thyroid Cancer
RUFY1 (x2) – Protein kinase domain retained – Thyroid Cancer
KIF5B (x2) – Protein kinase domain retained – Non-Small Cell Lung Cancer
CCDC6 (x2) – Protein kinase domain lost – Thyroid Cancer
KIF5B (x2) – Protein kinase domain lost – Non-Small Cell Lung Cancer
RUFY3 (x2) – Protein kinase domain retained – Thyroid Cancer
FBXL7 (x2) – Protein kinase domain retained – Non-Small Cell Lung Cancer
UXS1 – Protein kinase domain lost – Hepatobiliary Cancer
KIAA1217 – Protein kinase domain retained – Esophagogastric Cancer
KCNMA1 – Protein kinase domain retained – Esophagogastric Cancer
KIF5B – Protein kinase domain lost – Non-Small Cell Lung Cancer
KIF5B – Protein kinase domain retained – Non-Small Cell Lung Cancer
DOCK1 – Protein kinase domain retained – Non-Small Cell Lung Cancer
CCDC6 – Protein kinase domain retained – Esophagogastric Cancer
LIN52 – Protein kinase domain lost – Prostate Cancer
KIF5B – Protein kinase domain retained – Non-Small Cell Lung Cancer
CCDC6 – Protein kinase domain retained – Colorectal Cancer
CCDC6 – Protein kinase domain lost – Thyroid Cancer
PDCD10 – Protein kinase domain lost – Thyroid Cancer
TFG – Protein kinase domain retained – Thyroid Cancer
ERC1 – Protein kinase domain retained – Non-Small Cell Lung Cancer
CCDC6 – Protein kinase domain retained – Thyroid Cancer
RBPMS – Protein kinase domain retained – Non-Small Cell Lung Cancer

Partner-block colors are decorative; row labels identify partners.

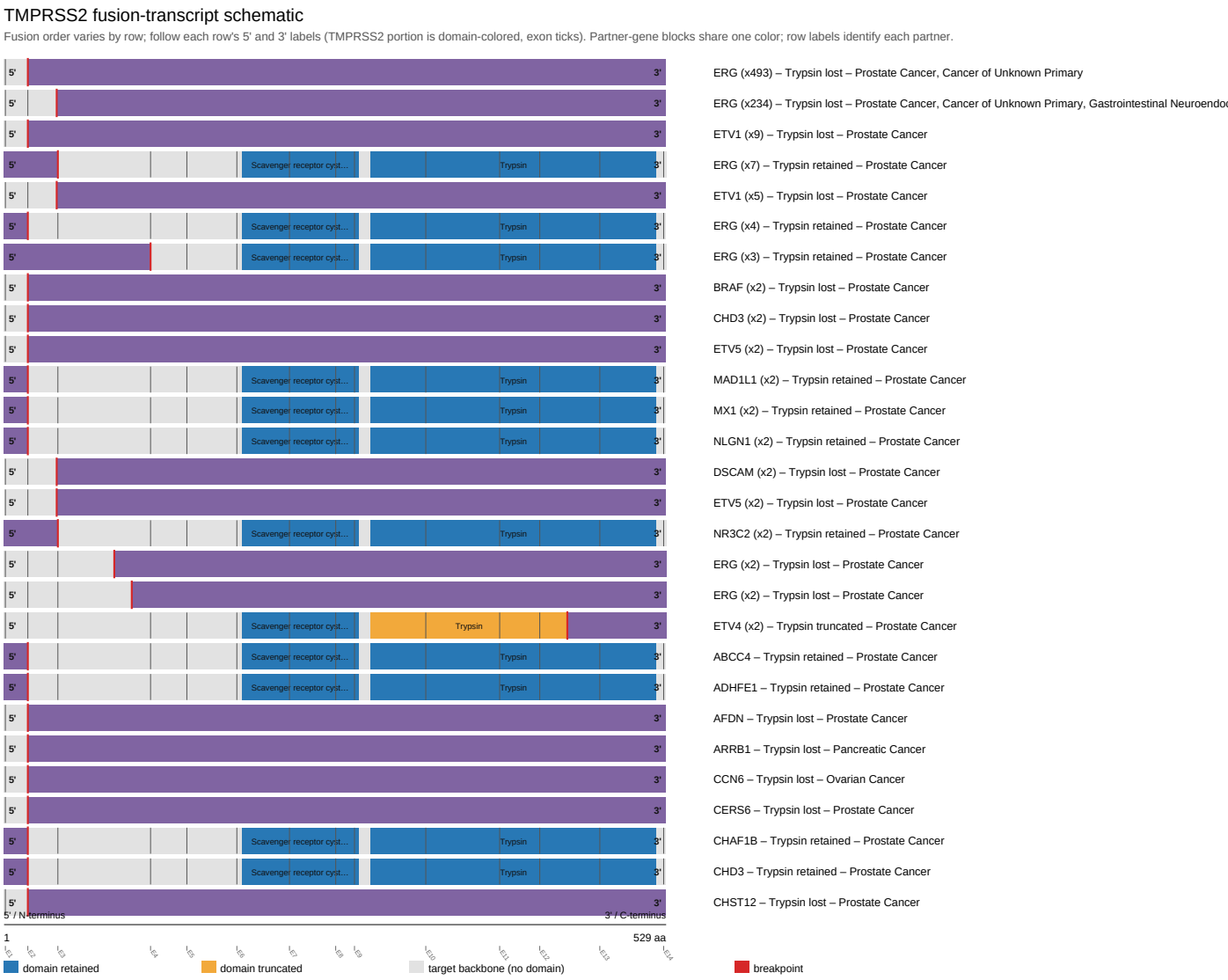
Showing the top 28 of 45 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/ret/report.md

TMPRSS2 (FDR-significant)

TMPRSS2 was analyzed across 867 fusion events, 33.3% in-frame and 6.2% domain-retained. Domain-retention Fisher's exact test on TMPRSS2's own fusion events alone gives p=0.0849513 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, TMPRSS2's Benjamini-Hochberg-adjusted q=0.000624279 (reaches genome-wide FDR significance).

Reused from the TMPRSS2 individual gene report (fusion_schematic.svg).

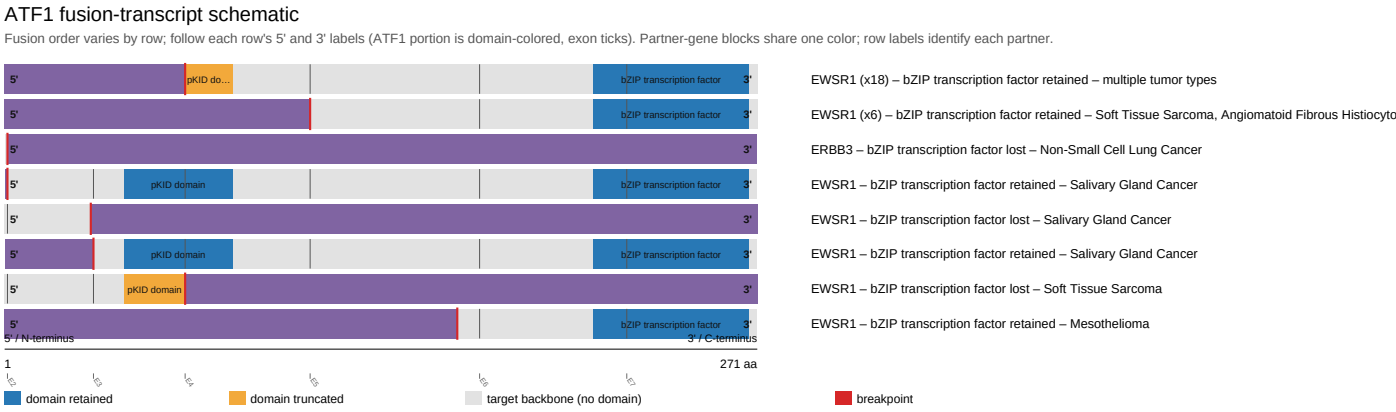


Full per-gene detail: gene_reports/tmprss2/report.md

ATF1 (FDR-significant)

ATF1 was analyzed across 30 fusion events, 84.0% in-frame and 90.0% domain-retained. Domain-retention Fisher's exact test on ATF1's own fusion events alone gives p=0.704212 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, ATF1's Benjamini-Hochberg-adjusted q=0.00109249 (reaches genome-wide FDR significance).

Reused from the ATF1 individual gene report (fusion_schematic.svg).

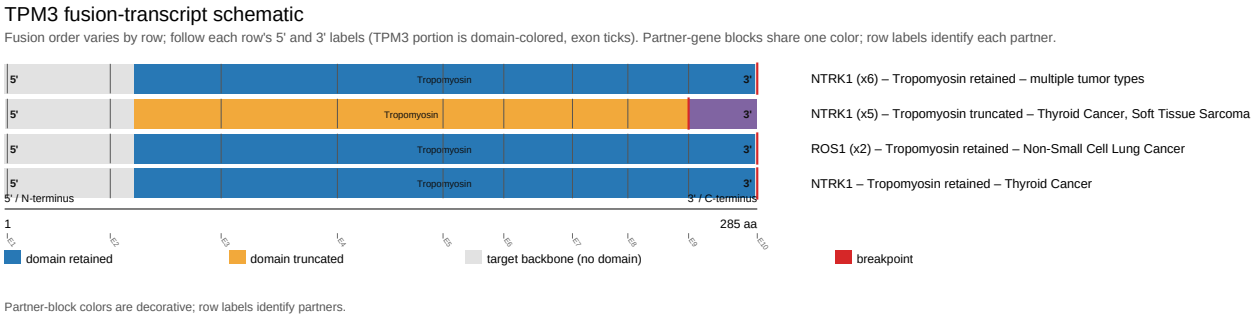


Full per-gene detail: gene_reports/atf1/report.md

TPM3 (FDR-significant)

TPM3 was analyzed across 14 fusion events, 100.0% in-frame and 64.3% domain-retained. Domain-retention Fisher's exact test on TPM3's own fusion events alone gives p=1 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, TPM3's Benjamini-Hochberg-adjusted q=0.00109249 (reaches genome-wide FDR significance).

Reused from the TPM3 individual gene report (fusion_schematic.svg).

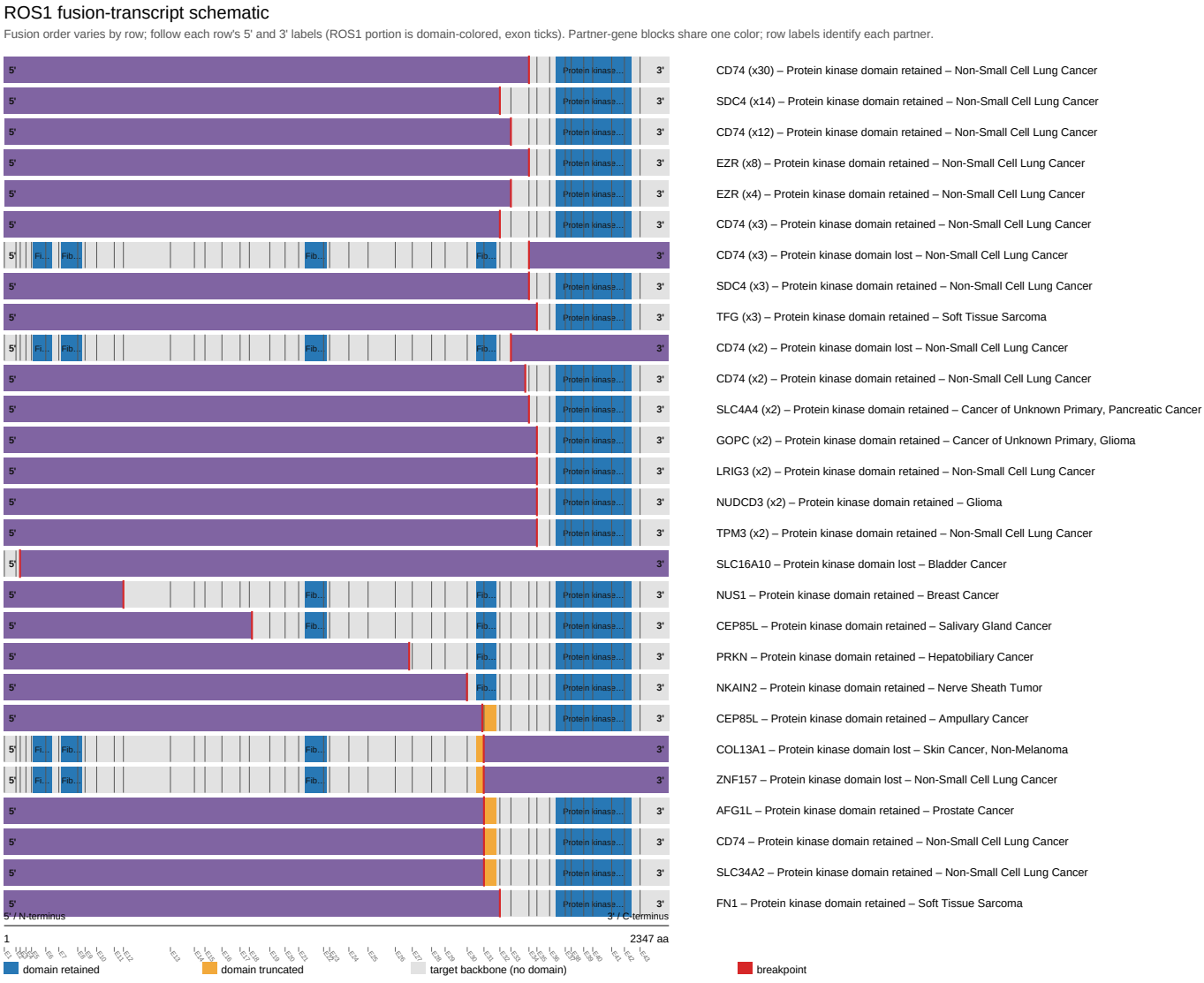


Full per-gene detail: gene_reports/tpm3/report.md

ROS1 (FDR-significant, Curated gene config)

ROS1 was analyzed across 122 fusion events, 72.7% in-frame and 90.7% domain-retained. Domain-retention Fisher's exact test on ROS1's own fusion events alone gives p=0.186019 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, ROS1's Benjamini-Hochberg-adjusted q=0.0019422 (reaches genome-wide FDR significance).

Reused from the ROS1 individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/ros1/report.md

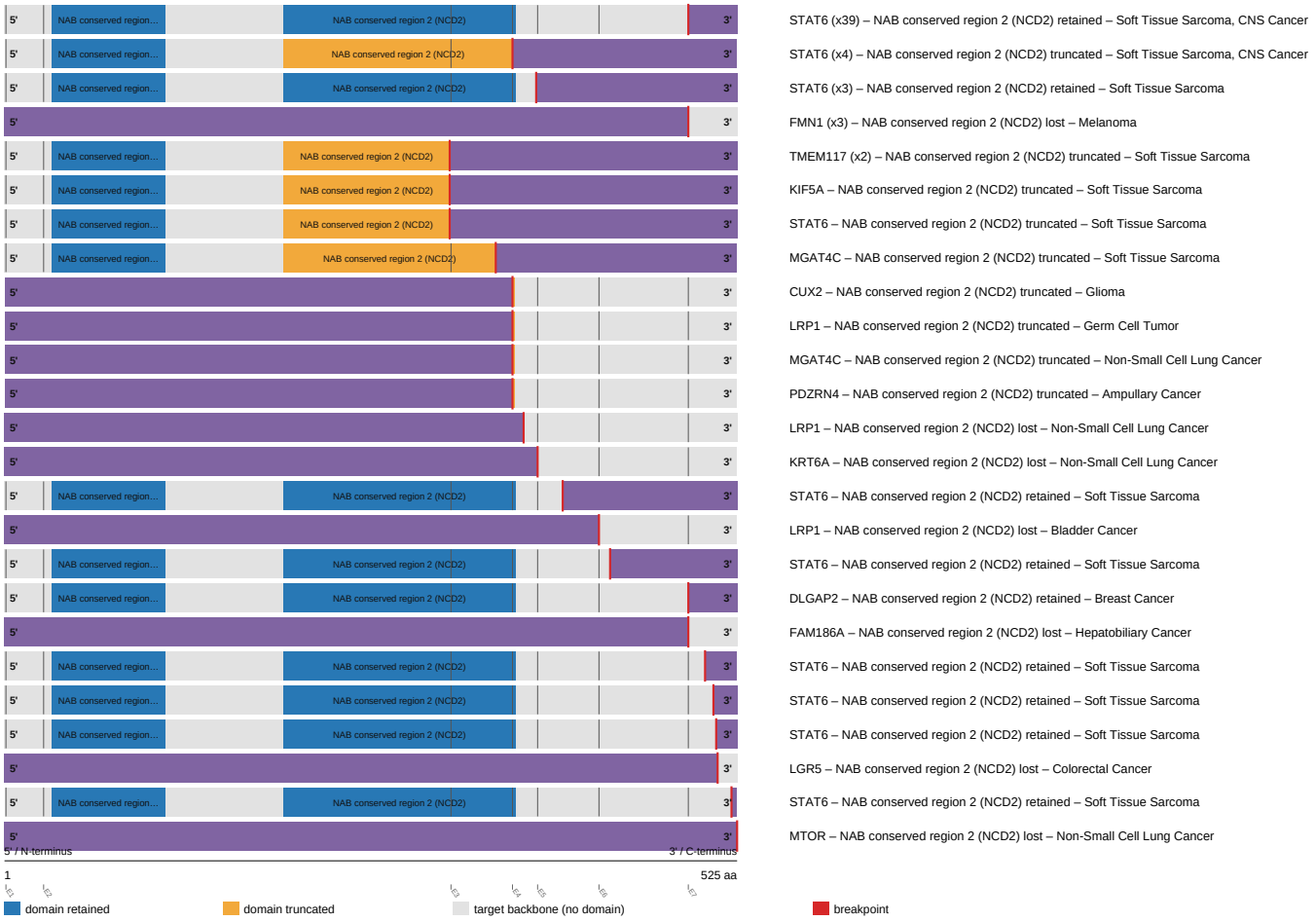
NAB2 (FDR-significant)

NAB2 was analyzed across 72 fusion events, 71.4% in-frame and 69.0% domain-retained. Domain-retention Fisher's exact test on NAB2's own fusion events alone gives p=0.179862 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, NAB2's Benjamini-Hochberg-adjusted q=0.00919908 (reaches genome-wide FDR significance).

Reused from the NAB2 individual gene report (fusion_schematic.svg).

NAB2 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (NAB2 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



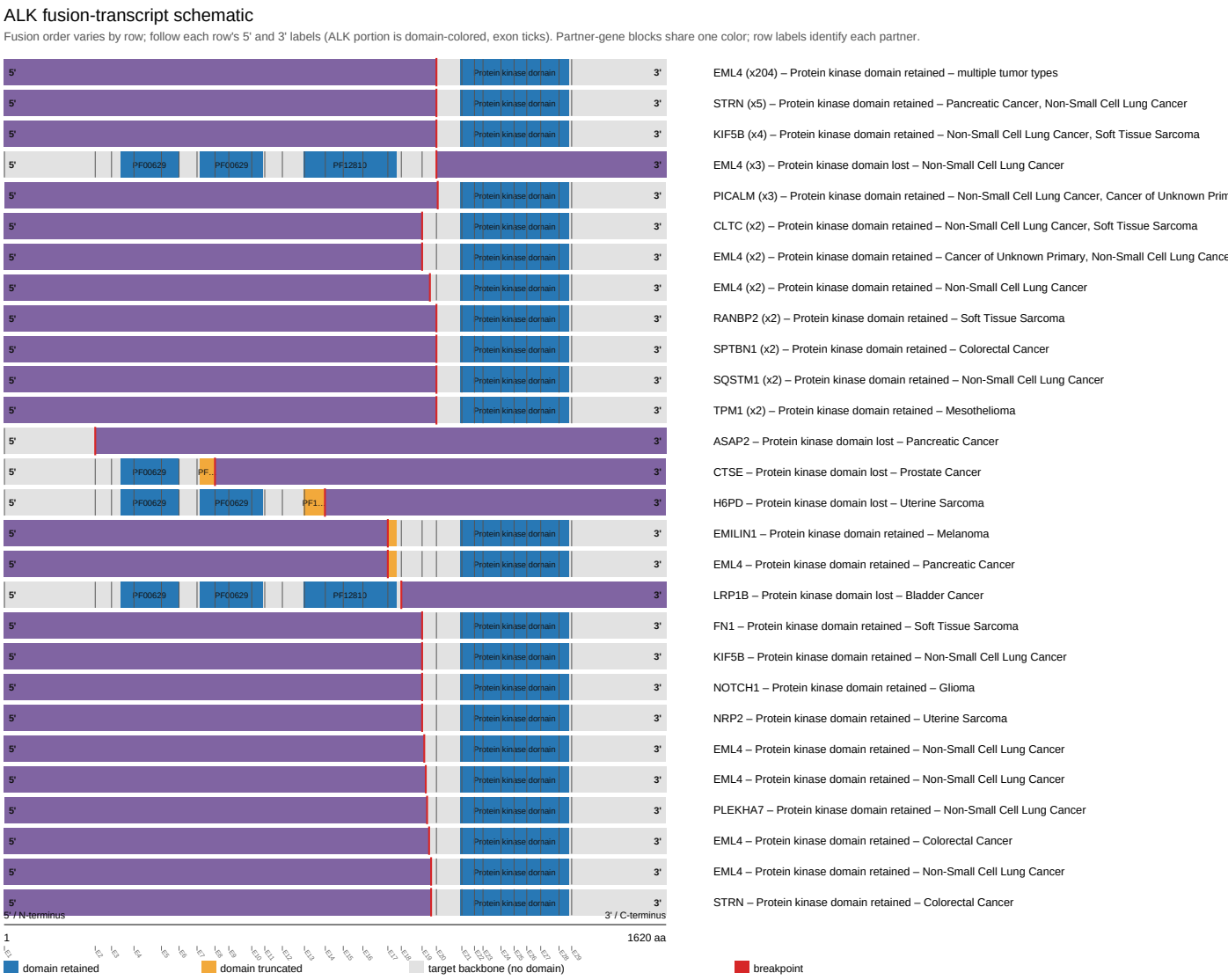
Partner-block colors are decorative; row labels identify partners.

Full per-gene detail: gene_reports/nab2/report.md

ALK (FDR-significant, Curated gene config)

ALK was analyzed across 272 fusion events, 97.4% in-frame and 96.3% domain-retained. Domain-retention Fisher's exact test on ALK's own fusion events alone gives p=0.00057792 (statistically significant at alpha=0.05). The Protein kinase domain appears to be required for retention. Among 285 genes scanned genome-wide in this cohort run, ALK's Benjamini-Hochberg-adjusted q=0.016646 (reaches genome-wide FDR significance).

Reused from the ALK individual gene report (fusion_schematic.svg).

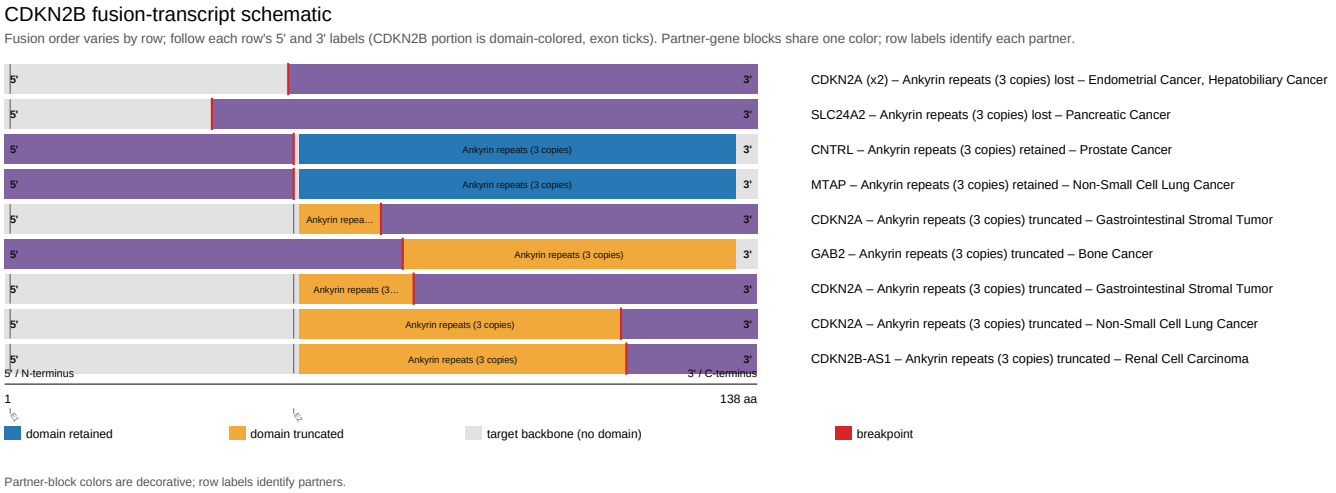


Full per-gene detail: gene_reports/alk/report.md

CDKN2B (Honorable mention)

CDKN2B was analyzed across 12 fusion events, 66.7% in-frame and 20.0% domain-retained. Domain-retention Fisher's exact test on CDKN2B's own fusion events alone gives p=0.0222222 (statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, CDKN2B's Benjamini-Hochberg-adjusted q=0.524925 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the CDKN2B individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/cdkn2b/report.md

NTRK1 (Honorable mention, Curated gene config)

NTRK1 was analyzed across 78 fusion events, 72.7% in-frame and 84.2% domain-retained. Domain-retention Fisher's exact test on NTRK1's own fusion events alone gives p=0.0514497 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, NTRK1's Benjamini-Hochberg-adjusted q=0.320147 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the NTRK1 individual gene report (fusion_schematic.svg).

NTRK1 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (NTRK1 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

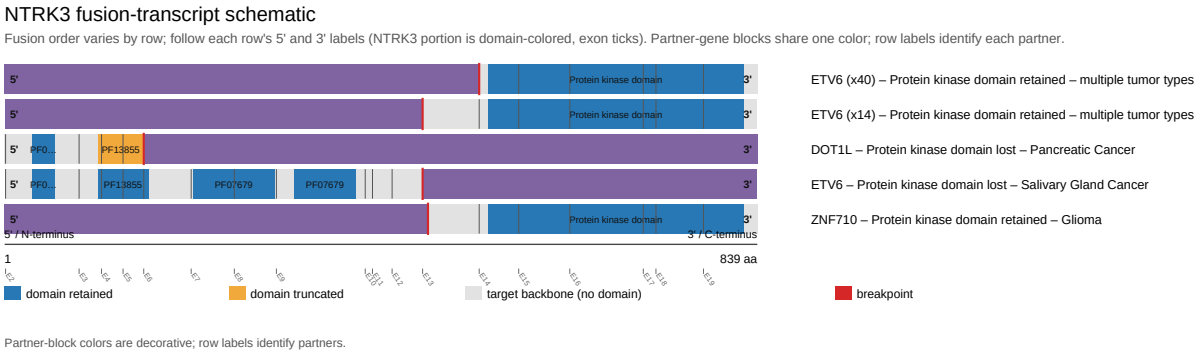
Showing the top 28 of 53 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/ntrk1/report.md

NTRK3 (Honorable mention, Curated gene config)

NTRK3 was analyzed across 58 fusion events, 98.2% in-frame and 96.5% domain-retained. Domain-retention Fisher's exact test on NTRK3's own fusion events alone gives p=0.0555556 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, NTRK3's Benjamini-Hochberg-adjusted q=0.0759241 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the NTRK3 individual gene report (fusion_schematic.svg).

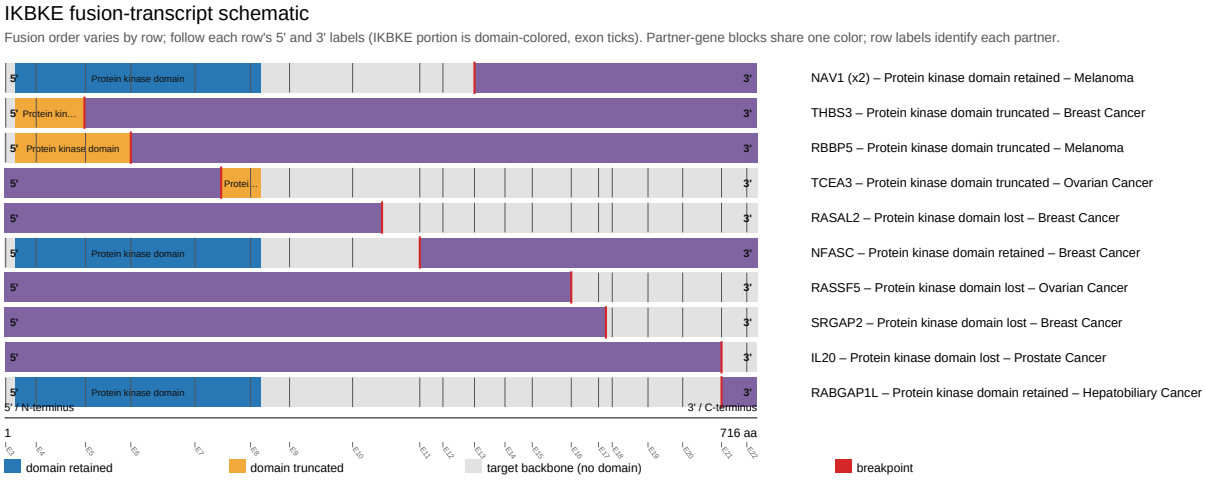


Full per-gene detail: gene_reports/ntrk3/report.md

IKBKE (Honorable mention)

IKBKE was analyzed across 12 fusion events, 50.0% in-frame and 36.4% domain-retained. Domain-retention Fisher's exact test on IKBKE's own fusion events alone gives p=0.0666667 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, IKBKE's Benjamini-Hochberg-adjusted q=1 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the IKBKE individual gene report (fusion_schematic.svg).

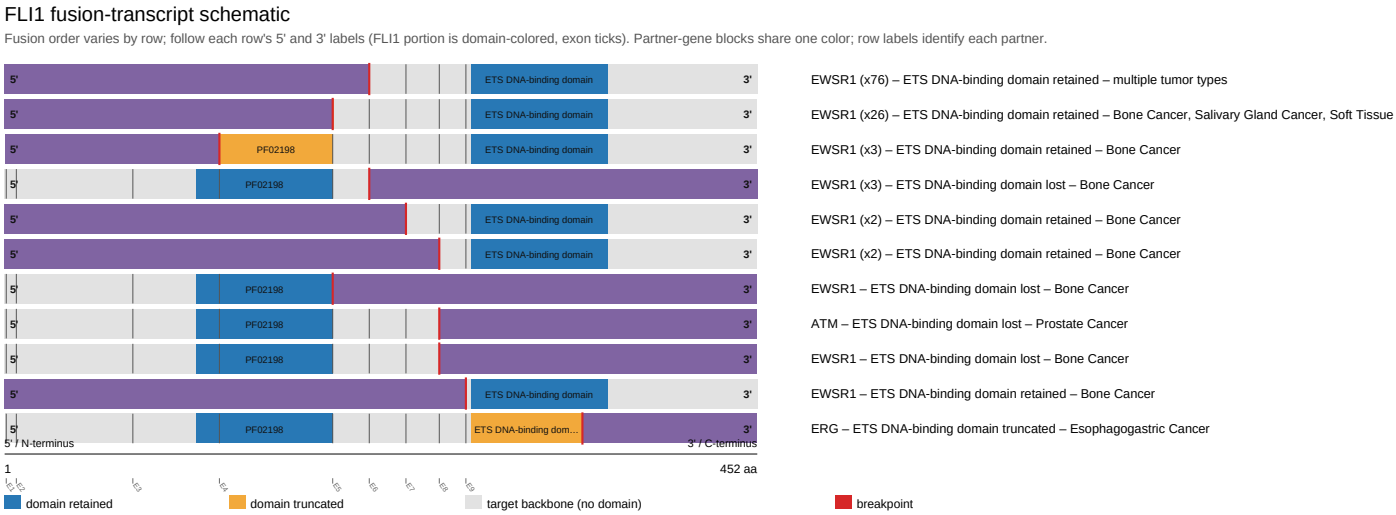


Full per-gene detail: gene_reports/ikbke/report.md

FLI1 (Honorable mention, Curated gene config)

FLI1 was analyzed across 118 fusion events, 68.2% in-frame and 94.0% domain-retained. Domain-retention Fisher's exact test on FLI1's own fusion events alone gives p=0.0768769 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, FLI1's Benjamini-Hochberg-adjusted q=1 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the FLI1 individual gene report (fusion_schematic.svg).

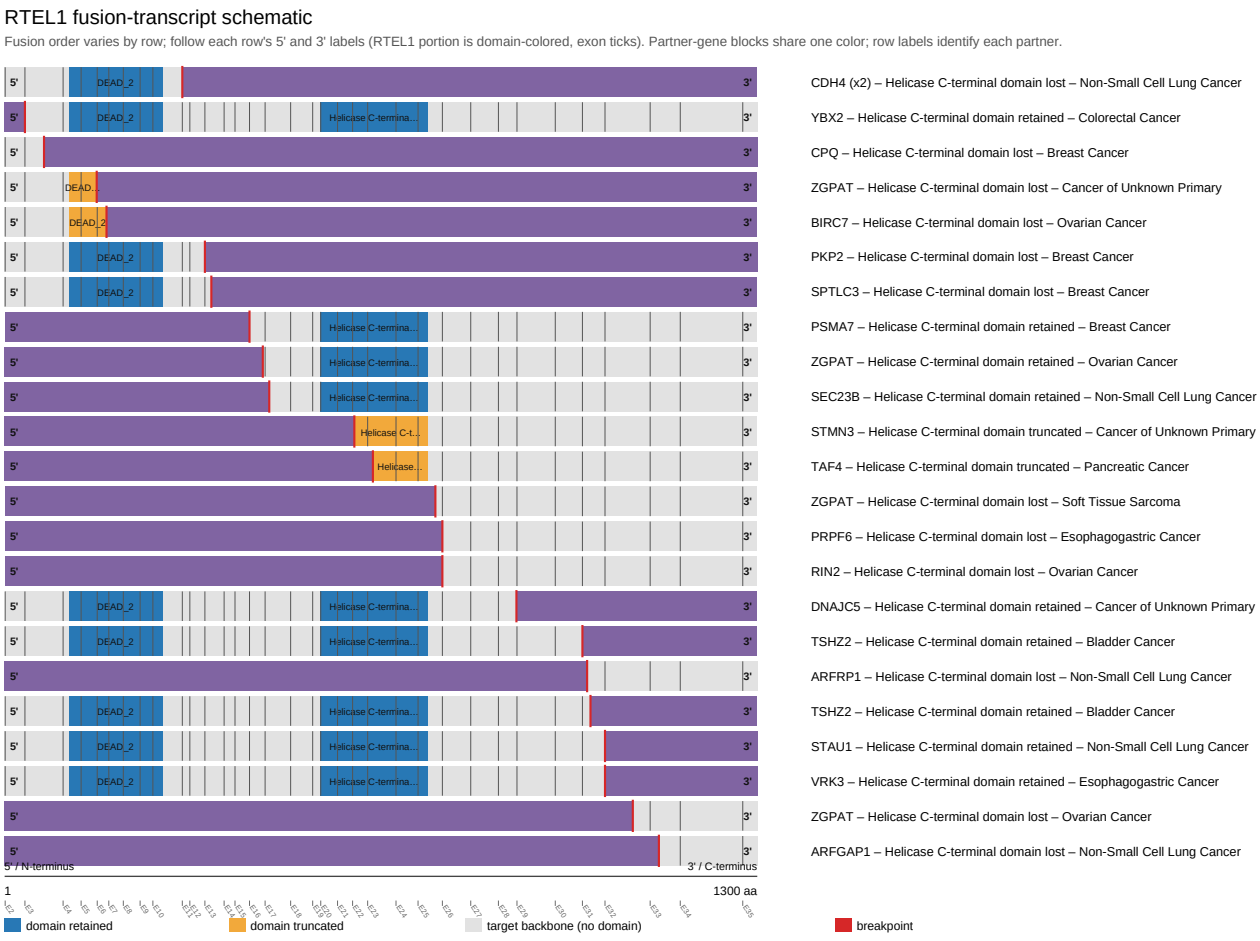


Full per-gene detail: gene_reports/fli1/report.md

RTEL1 (Honorable mention)

RTEL1 was analyzed across 24 fusion events, 50.0% in-frame and 37.5% domain-retained. Domain-retention Fisher's exact test on RTEL1's own fusion events alone gives p=0.116746 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, RTEL1's Benjamini-Hochberg-adjusted q=1 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the RTEL1 individual gene report (fusion_schematic.svg).

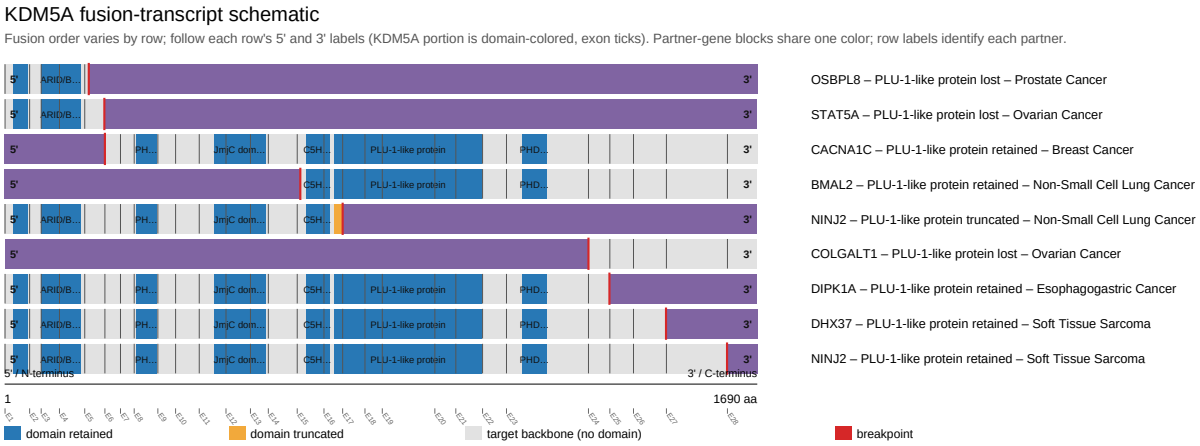


Full per-gene detail: gene_reports/rtel1/report.md

KDM5A (Honorable mention)

KDM5A was analyzed across 10 fusion events, 50.0% in-frame and 55.6% domain-retained. Domain-retention Fisher's exact test on KDM5A's own fusion events alone gives p=0.119048 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, KDM5A's Benjamini-Hochberg-adjusted q=1 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the KDM5A individual gene report (fusion_schematic.svg).

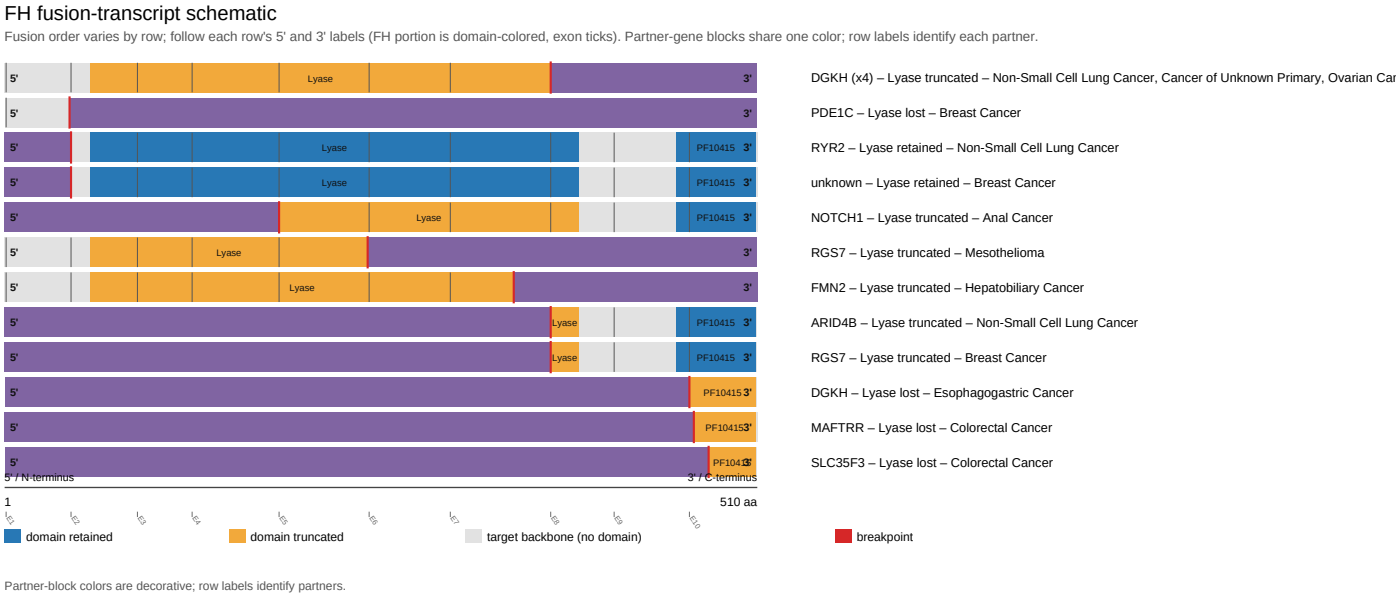


Full per-gene detail: gene_reports/kdm5a/report.md

FH (Honorable mention)

FH was analyzed across 16 fusion events, 16.7% in-frame and 13.3% domain-retained. Domain-retention Fisher's exact test on FH's own fusion events alone gives p=0.142857 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, FH's Benjamini-Hochberg-adjusted q=1 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the FH individual gene report (fusion_schematic.svg).



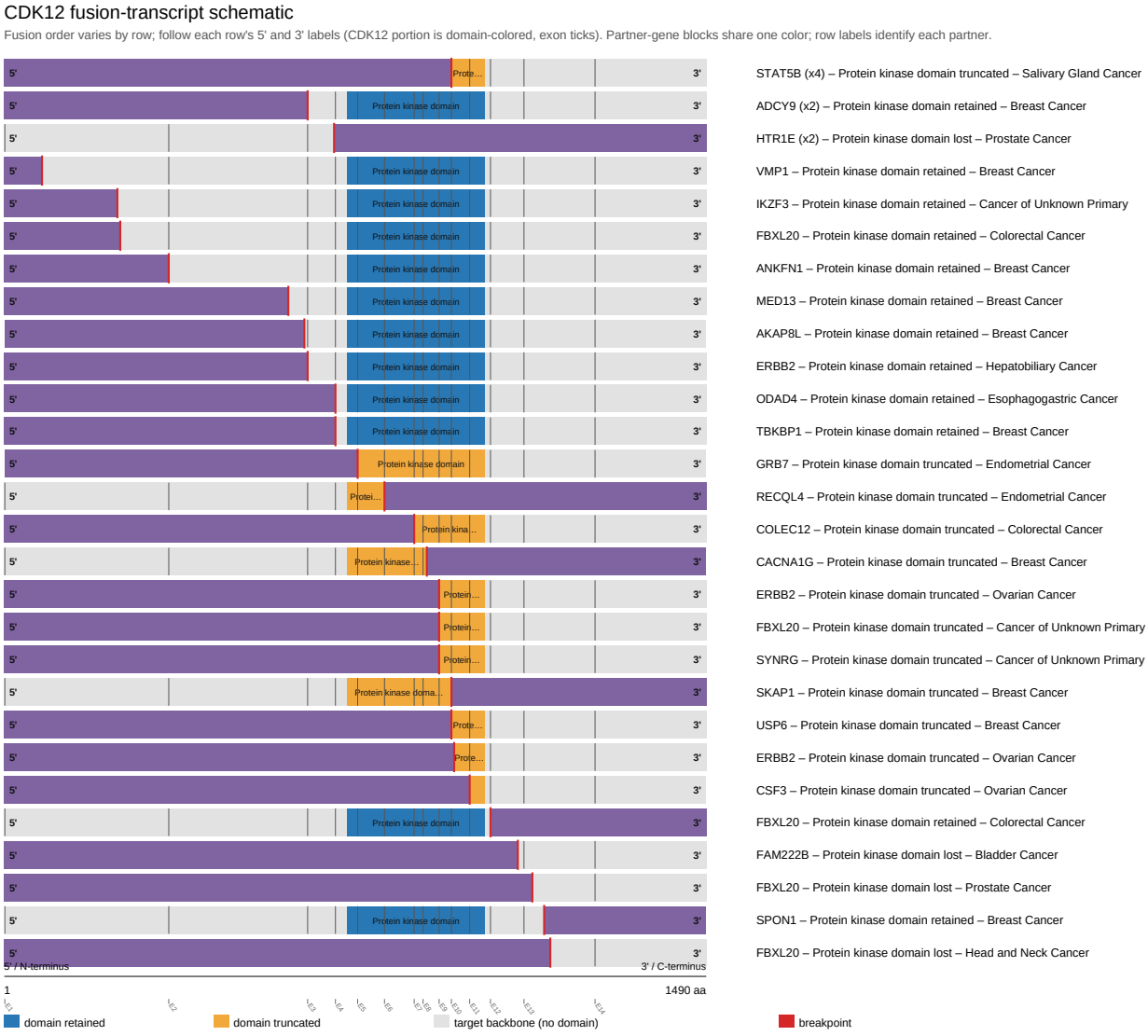
Full per-gene detail: gene_reports/fh/report.md

CDK12 (Honorable mention)

CDK12 was analyzed across 43 fusion events, 50.0% in-frame and 39.0% domain-retained. Domain-retention Fisher's exact test on CDK12's own fusion events alone gives p=0.17342 (not statistically significant at

alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, CDK12's Benjamini-Hochberg-adjusted q=1 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the CDK12 individual gene report (fusion_schematic.svg).



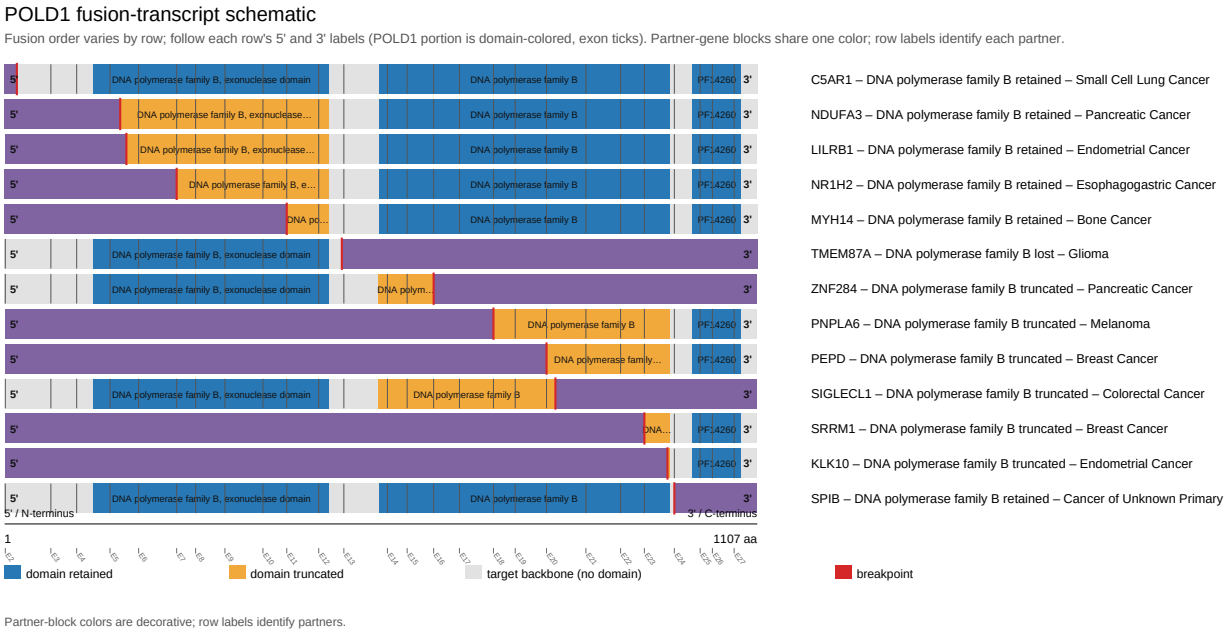
Full per-gene detail: gene_reports/cdk12/report.md

POLD1 (Honorable mention)

POLD1 was analyzed across 16 fusion events, 33.3% in-frame and 46.2% domain-retained. Domain-retention Fisher's exact test on POLD1's own fusion events alone gives p=0.192308 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, POLD1's Benjamini-Hochberg-adjusted q=0.735265 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is

NOT a claim of statistical significance.

Reused from the POLD1 individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/pold1/report.md

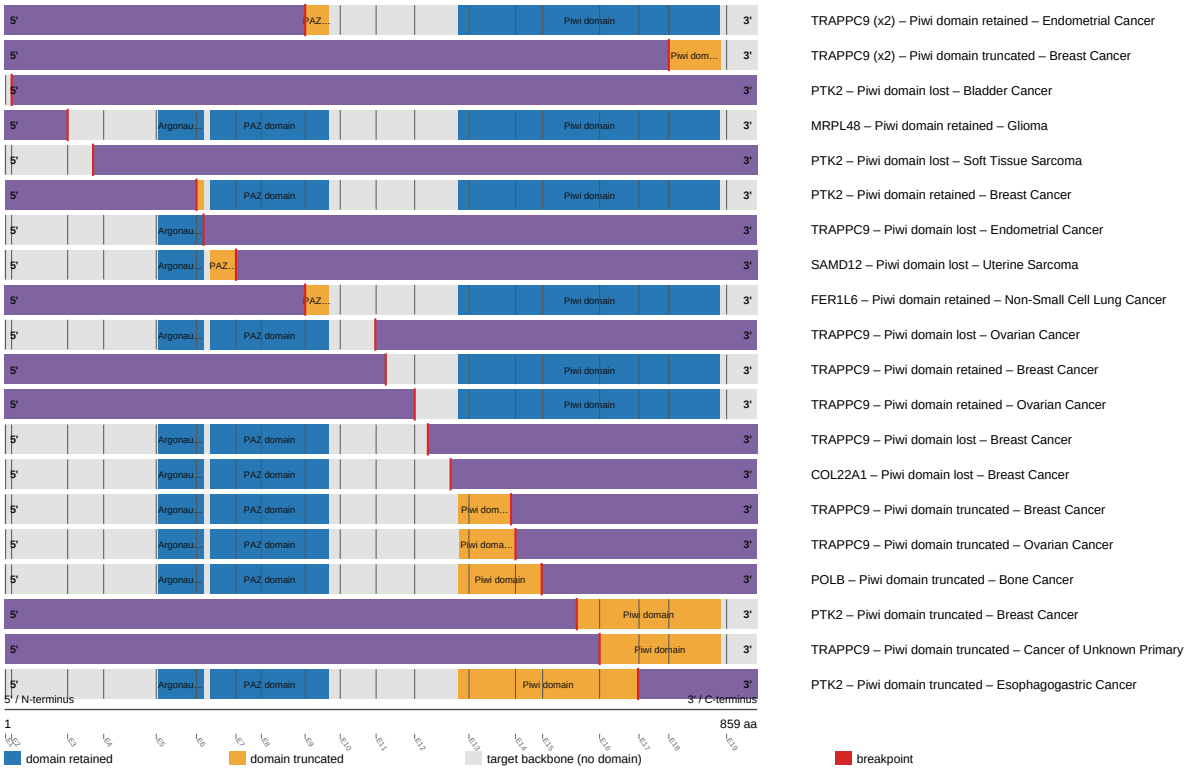
AGO2 (Honorable mention)

AGO2 was analyzed across 23 fusion events, 46.2% in-frame and 31.8% domain-retained. Domain-retention Fisher's exact test on AGO2's own fusion events alone gives p=0.225232 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, AGO2's Benjamini-Hochberg-adjusted q=1 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the AGO2 individual gene report (fusion_schematic.svg).

AGO2 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (AGO2 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

Full per-gene detail: gene_reports/ago2/report.md

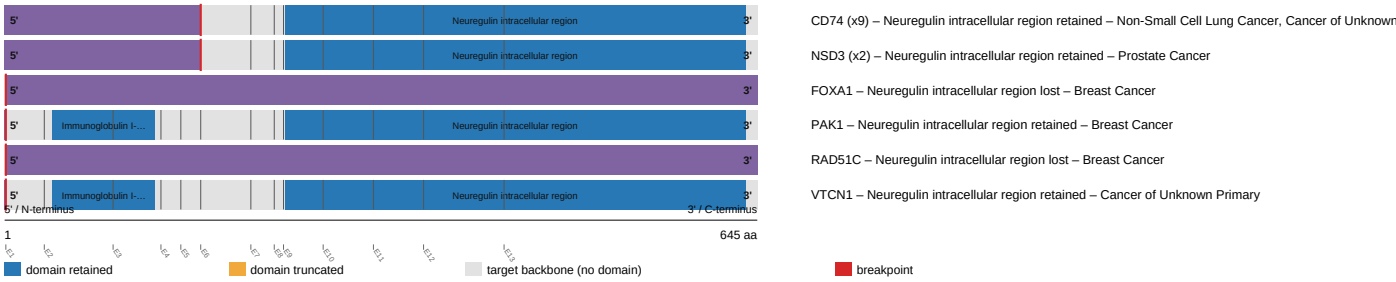
NRG1 (Honorable mention)

NRG1 was analyzed across 15 fusion events, 100.0% in-frame and 86.7% domain-retained. Domain-retention Fisher's exact test on NRG1's own fusion events alone gives p=0.227273 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, NRG1's Benjamini-Hochberg-adjusted q=1 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the NRG1 individual gene report (fusion_schematic.svg).

NRG1 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (NRG1 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



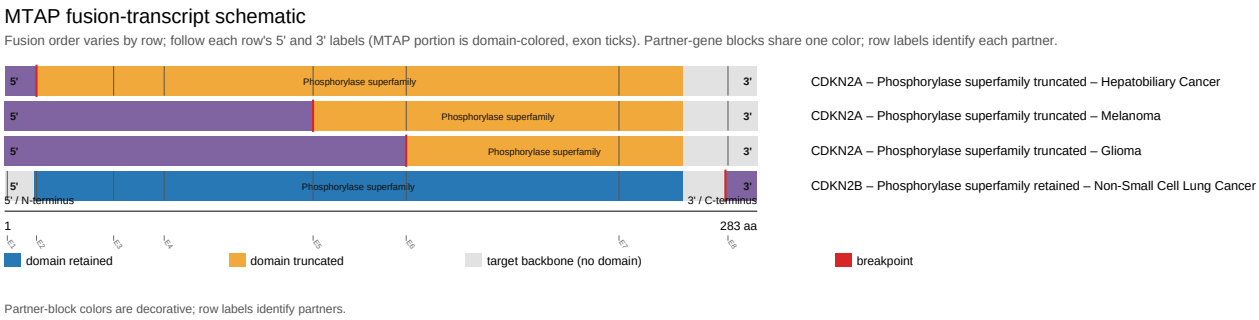
Partner-block colors are decorative; row labels identify partners.

Full per-gene detail: gene_reports/nrg1/report.md

MTAP (Honorable mention)

MTAP was analyzed across 4 fusion events, 33.3% in-frame and 25.0% domain-retained. Domain-retention Fisher's exact test on MTAP's own fusion events alone gives p=0.25 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, MTAP's Benjamini-Hochberg-adjusted q=1 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the MTAP individual gene report (fusion_schematic.svg).

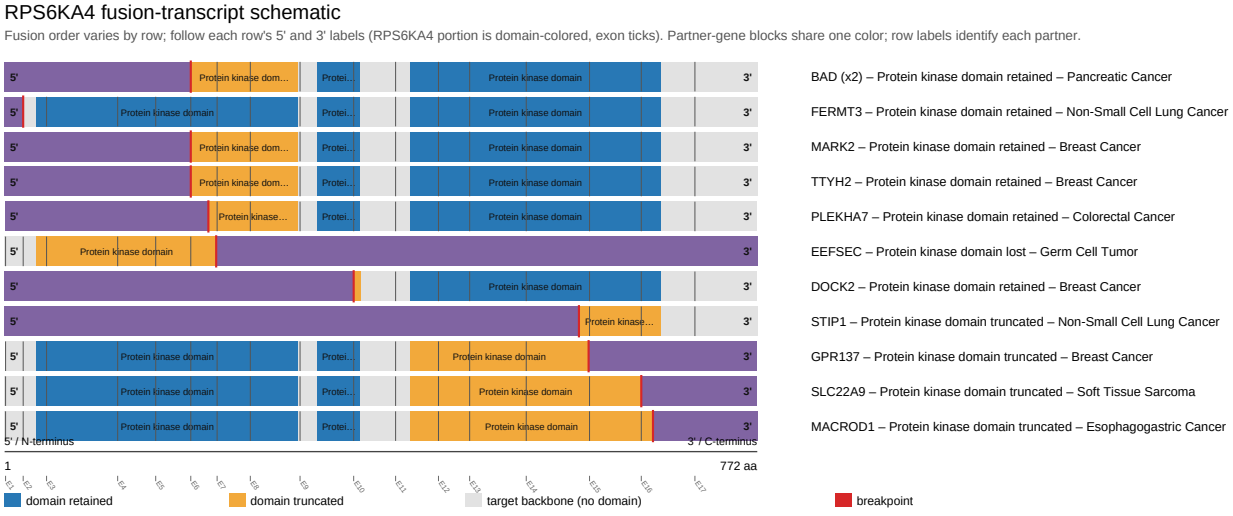


Full per-gene detail: gene_reports/mtap/report.md

RPS6KA4 (Honorable mention)

RPS6KA4 was analyzed across 13 fusion events, 62.5% in-frame and 58.3% domain-retained. Domain-retention Fisher's exact test on RPS6KA4's own fusion events alone gives p=0.261905 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, RPS6KA4's Benjamini-Hochberg-adjusted q=1 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the RPS6KA4 individual gene report (fusion_schematic.svg).

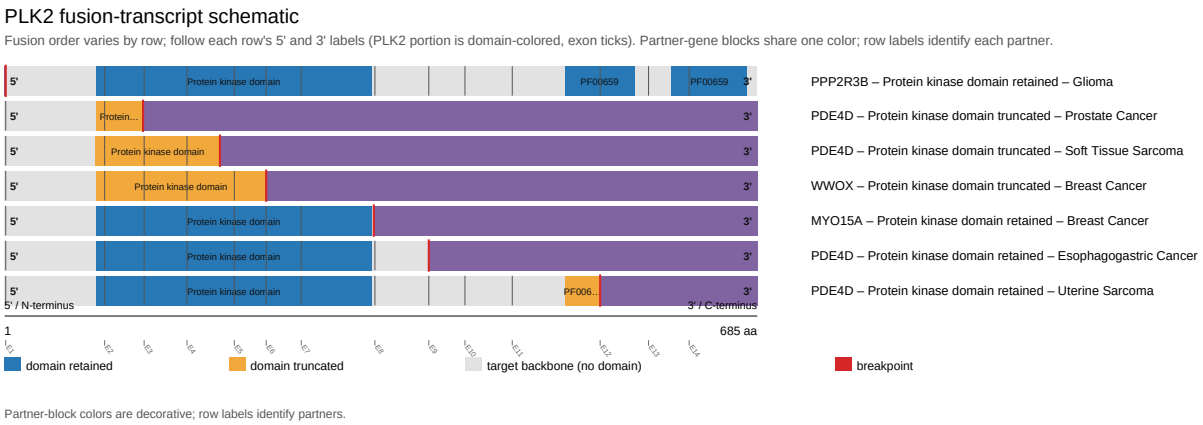


Full per-gene detail: gene_reports/rps6ka4/report.md

PLK2 (Honorable mention)

PLK2 was analyzed across 7 fusion events, 50.0% in-frame and 57.1% domain-retained. Domain-retention Fisher's exact test on PLK2's own fusion events alone gives p=0.285714 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, PLK2's Benjamini-Hochberg-adjusted q=1 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the PLK2 individual gene report (fusion_schematic.svg).

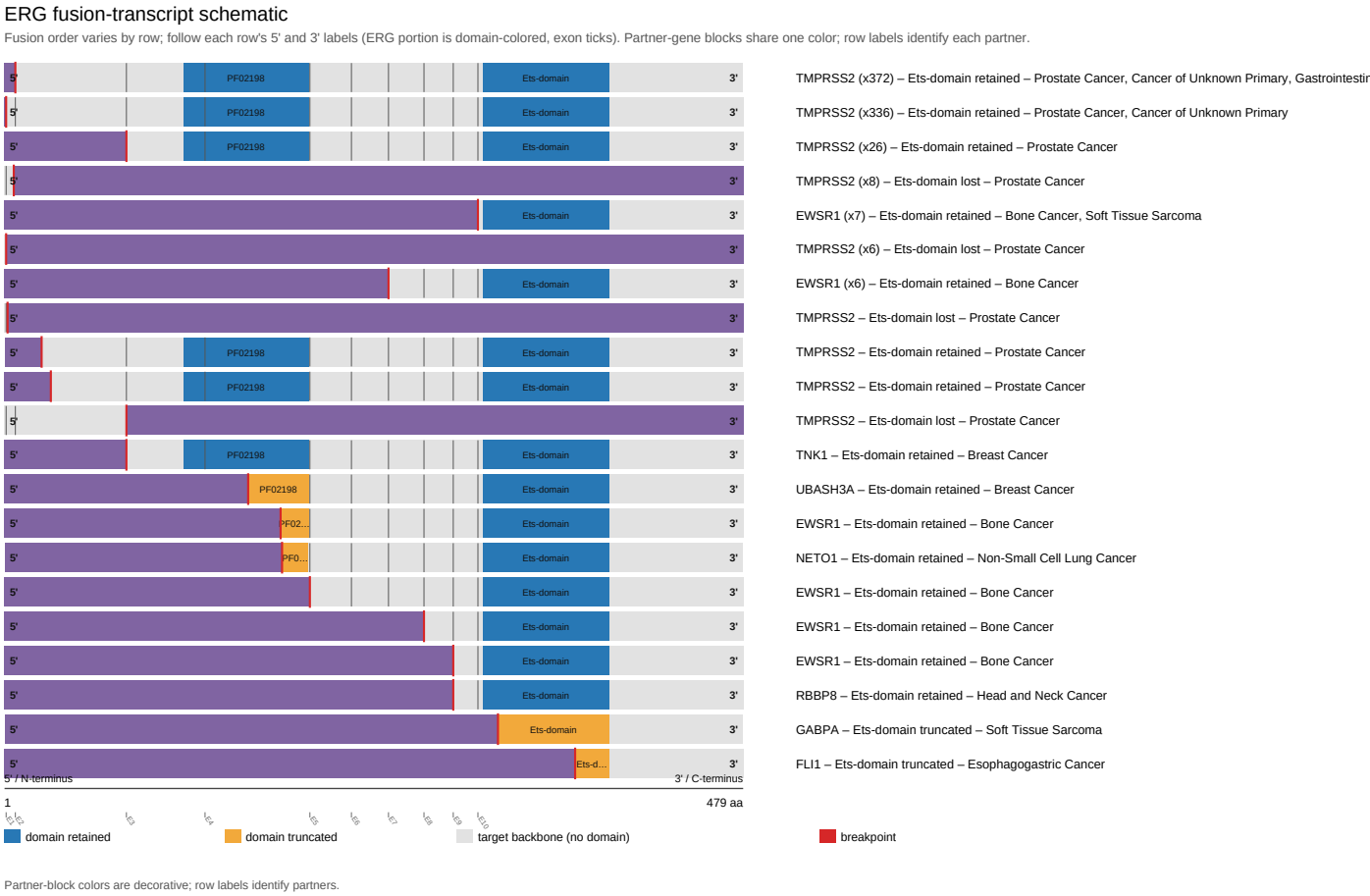


Full per-gene detail: gene_reports/plk2/report.md

ERG (Curated gene config)

ERG was analyzed across 788 fusion events, 33.0% in-frame and 97.7% domain-retained. Domain-retention Fisher's exact test on ERG's own fusion events alone gives p=0.778543 (not statistically significant at alpha=0.05). Among 285 genes scanned genome-wide in this cohort run, ERG's Benjamini-Hochberg-adjusted q=0.100745 (does not reach genome-wide FDR significance).

Reused from the ERG individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/erg/report.md

Discussion

• Cross-gene Benjamini-Hochberg FDR correction was applied jointly across the 285 genes that produced at least one computable p-value in this scan. This reduces false-positive findings relative to testing each gene in isolation, but is a conservative correction: a real per-gene effect can fail to reach the q < 0.05 threshold once corrected across that full p-value-bearing gene set.

• 10 gene(s) used hand-curated gene configurations, while 528 gene(s) were auto-configured from Genome Nexus canonical-transcript/Pfam-domain data using a kinase/catalytic-keyword heuristic to select the tracked domain; auto-configured domains have not been manually verified the way hand-curated ones have.

• Data were retrieved live from the public msk_impact_50k_2026 cBioPortal study as of 2026-10-29T15:52:29.772745+00:00. Some cBioPortal cohorts (e.g. actively accruing clinical-sequencing panels such as MSK-IMPACT) are updated periodically, so exact counts could shift if this scan is re-run later against such a cohort.

• All findings in this manuscript are computational, hypothesis-generating candidate evidence from bioinformatic analysis of public cohort data, not validated clinical calls; a gene's presence here is not a therapeutic or diagnostic recommendation.

Appendix: per-gene report index

Gene	Tier	Config source	N events	Best q	Full report
BRAF	FDR-significant; Curated gene config	curated	179	0.0006243	gene_reports/braf/report.md
CD74	FDR-significant	auto	67	0.0006243	gene_reports/cd74/report.md
EGFR	FDR-significant	auto	55	0.0006243	gene_reports/egfr/report.md
EML4	FDR-significant	auto	223	0.0006243	gene_reports/eml4/report.md
ETV6	FDR-significant; Curated gene config	curated	90	0.0006243	gene_reports/etv6/report.md
FGFR2	FDR-significant; Curated gene config	curated	136	0.0006243	gene_reports/fgfr2/report.md
FGFR3	FDR-significant	auto	152	0.0006243	gene_reports/fgfr3/report.md
KIAA1549	FDR-significant	auto	44	0.0006243	gene_reports/kiaa1549/report.md
KIF5B	FDR-significant	auto	96	0.0006243	gene_reports/kif5b/report.md
PRKACA	FDR-significant	auto	35	0.0006243	gene_reports/prkaca/report.md
RET	FDR-significant; Curated gene config	curated	194	0.0006243	gene_reports/ret/report.md
TMPRSS2	FDR-significant	auto	867	0.0006243	gene_reports/tmprss2/report.md
ATF1	FDR-significant	auto	30	0.001092	gene_reports/atf1/report.md
TPM3	FDR-significant	auto	14	0.001092	gene_reports/tpm3/report.md
ROS1	FDR-significant; Curated gene config	curated	122	0.001942	gene_reports/ros1/report.md
NAB2	FDR-significant	auto	72	0.009199	gene_reports/nab2/report.md
ALK	FDR-significant; Curated gene config	curated	272	0.01665	gene_reports/alk/report.md
CDKN2B	Honorable mention	auto	12	0.5249	gene_reports/cdkn2b/report.md
NTRK1	Honorable mention; Curated gene config	curated	78	0.3201	gene_reports/ntrk1/report.md
NTRK3	Honorable mention; Curated gene config	curated	58	0.07592	gene_reports/ntrk3/report.md
IKBKE	Honorable mention	auto	12	1	gene_reports/ikbke/report.md
FLI1	Honorable mention; Curated gene config	curated	118	1	gene_reports/fli1/report.md
RTEL1	Honorable mention	auto	24	1	gene_reports/rtel1/report.md
KDM5A	Honorable mention	auto	10	1	gene_reports/kdm5a/report.md
FH	Honorable mention	auto	16	1	gene_reports/fh/report.md
CDK12	Honorable mention	auto	43	1	gene_reports/cdk12/report.md
POLD1	Honorable mention	auto	16	0.7353	gene_reports/pold1/report.md
AGO2	Honorable mention	auto	23	1	gene_reports/ago2/report.md
NRG1	Honorable mention	auto	15	1	gene_reports/nrg1/report.md
MTAP	Honorable mention	auto	4	1	gene_reports/mtap/report.md
RPS6KA4	Honorable mention	auto	13	1	gene_reports/rps6ka4/report.md
PLK2	Honorable mention	auto	7	1	gene_reports/plk2/report.md
ERG	Curated gene config	curated	788	0.1007	gene_reports/erg/report.md